# Subject: Bio301 (Notes)

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# VU Medical Zone (biotechnologists)

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1) Population genetics is the study of_____in a group of population.?

a) gene b) heredity c) trait d) genome

2) Quantitative genetics is the study of group hereditary for traits determined by many genes simultaneously such as____?

a) skin color, b) height, c)eye color d) all of these

3) _____is the branch of genetics deals with the molecular structure and function of genes

a) Molecular genetics b) heredity genetic c)chemical genetic d) genotyping

4)When one characteristic expresses itself over the other i.e. round over wrinkled was dominant in Gregor Mendel experiments IS Called ?

a) Qualitative trait b) Recessive traitc) Dominant trait d) quantitative trait

5) if a trait does not show through in the first generation is called as_____ i.e. wrinkled.?

a) Qualitative trait b) Recessive trait c) Dominant trait d) quantitative trait

6) When we cross parents, that are pure for contrasting traits, if only one form of the trait will appear in the next generation. In the monohybrid cross, one version will be disappeared is called_____?

a) lawndependent assortment b) law of dominance c) Law of Segregation

7)when Indicate "a change" while in other disciplines it is used to indicate "a disease causes_____change" ?

a) Mendelian genetics b) Non-Mendelian genetics c) gene d) mutation

8)in which of the following the Genetic make-up of an organism held ?

a) hereditb) Genotype c) phenotype d) gene

9) To create physical appearance of an organism is called_____?

a) heredity b) Genotype c) phenotype d) gene

10 The process of Transmission of traits from parents to offspring is called____?

a) trait b) gene c) papulationd) Heredity

11) According to blending hypothesis, genetic material of the parents mixes during_____ ?

a) egg b) ovum c) fertilization d) sperm

12) the process in which Differences can be seen in individuals of the same species, such as height, weight, color etc is called_____ ?

a) variation b) Discontinuous Variation c) Continuous variation d)heredity variation

13) Language, obesity, athletic skills etc are what type of variation?

a) Inherited Variations b) Acquired Variation c) Variation by genetic d)environmental variation

14) Blood groups, finger prints and sex etc arewhat type of ofvaration ?

a) Inherited Variations b) Acquired Variation c) Variation by genetic d)environmental variation

15) In which of the following process we describes the situation in which there are many intermediates between the extremes.

a) Variation b) Discontinuous Variation c) Continuous variation d) heredity variation

16) The process we are entirely genetically controlled. And it cannot be altered by external conditions. Blood group, color blind, dwarfism are genetically controlled are called?

A) Discontinuous Variation b) Acquired Variation c) Variation by genetic d) Continuous variation

17) Which process in which tall plants produced only tall offspring & purple flowered plants produced only purple are_____?

a) Pure Cross b) Pure-breeding plants c) hybrid cross d) all of these

18) Which of the following is aPattern of inheritance thatdeviates from Medellin pattern of inheritance is called ?

a) Medellin inheritance b) Non-Mendelian inheritance c) both of these d) none of these

**19) Each trait is controlled by_____gene?**

**a) DOUBLE gene b) triple gene c) double pair of gene d) single gene**

20) Each gene has_____alleles?

a) 3 b) 2 c) 5 d) 1

21) Cross of true-breeding strains resemble only of the parents?

a) One b) two c) three d) four

22) If two members of a gene pair segregate (separate) from each other during the formation of gametes is called_____?

a) law of dominanceb) Law of segregation c) Law of independent d) none of these

23) Some alleles for a gene are not completely dominant are called_____?

a) Complete dominance b) Incomplete dominance c) both of these

24) A situation, in which both alleles appear in the phenotype is called_____ ?

a) Co-dominance b)) complete dominance c) incomplete dominance d) none of these

25) Most traits are not controlled by a single gene locus, but by the combined interaction of many gene loci is called_____?

a) Qualitative trait b) quantitative trait c) Polygenic traitsd) none of these

26) Skin color pigmentation in humans is controlled by at least separately inherited genes?

a) 2 b) 3 c) 4 d) 6

27) Alleles at one gene locus can hide or prevent expression of alleles at second gene locus is called_____ ?

a) Polygenic trait b) variation c) Pleiotropd) Epistasis

28) The ability of a gene to affect an organism in many ways is called ?

a) Polygenic trait b) variation c) Pleiotrop d) Epistasis

29) Pleiotropy refers to an allele which has more than one effect on the ?

a) Genotype b) trait c) alleled) phenotype

30) Example of Sickle cell anemia and Cystic fibrosis is_____?

a) Polygenic trait b) variation c) Epistasis c) Pleiotro

31) In Huntington disease the Phenotype 1: progressive dementia at age of_____years?

a) 40-50 b) 50 -60 c) 50 -70 d) 50- 80

32) In Huntington disease the Phenotype 2: death in about _____after onset of disease. ?

a) 6 yearsb) 5 years c) 7 years d) 10 years

33) Chromosomes are the_____filamentous bodies present in the nucleus.

a) Circular shaped b) viral shaped c) squareshaped d) rod-shaped

34) Chromosomes were described by Strausberger in_____ ?

a)1875 b) 1877 c) 1870 d) 1879

35) Chromosome was first used by_____in 1888.

a) Strausberger b) Waldeye c) Robert Koch d) Elaxander Shelly

36) What are the meaning of wordChromosomes in biology? Without option

Ans; (Chromo = color; Soma = body).

37) Chromatin fibers undergo folding, coiling and supercoiling during_____so that the chromosomes become progressively thicker and smaller and become visible.

a) Metaphase b) anaphase c) prophase d) none of these

38) Chromosomes are rod shaped filamentous bodies present in _____cells.

a) Prokaryotic b) eukaryotic c) both of these d) none of these

39) Most organisms have chromosomes numbers between ?

1) a) 12 to 60 b) 12 to 50. C) 12 to 70 d) 12 to 80

40) Fungi have_____chromosomes?

a) 3-8 b) 3-9 c) 3-5 d) 3-3

41) Diploid chromosome are _____in number in human body.

a) 48 b) 46 c) 40 d) 42

42) How many Chromosomesare there in_____Mosquito body.

a) 8 b) 2 c) 4 d) 6

43) How many Chromosomes are there in_____Horsetail.

a) 200 b) 216 c) 220 d) 215

44.) The size of the chromosomes varies between 0.5 µ to 32 µ in length

a) 0.5 µ to 32 µ b) 0.5 µ to 30 µ c) 0.6 µ to 32 µ d) 0.6 µ to 30 µ

45) Haploids are able to express both recessive and dominant alleles/mutations because there are_____?

a) Many alleles for each gene b) two alleles for each gene

(c) One allele for each gene in the individual (d) only one allele in a gene.

46) A family of five daughters only is expecting sixth issue. The chance of its beings a son is_____%?
(a) zero(b) 25%(c) 50%(d) 100%

47) Mutations used in agriculture are commonly_____?
(a) induced(b) spontaneous(c) lethal(d) recessive and lethal
48) DNA replication is_____
(a) conservative and discontinuous(b) semiconservative and semidiscontinuous

(c) semiconservative and discontinuous(d) conservative.

49)Anoctamer of 4 histones complexed with DNA forms____?
(a) endosome(b) nucleosome(c) mesosome(d) centromere

49) Bateson used the terms coupling and repulsion for linkage and crossing over. Name the correct parental of coupling type along with its cross over or repulsion

a) coupling AABB, aabb; Repulsion AABB, aabb

(b) couplingAAbb, aaBB; Repulsion AaBb, aabb

(c) couplingaaBB, aabb; Repulsion A ABB, aabb

(d) coupling AABB, aabb : Repulsion AAbb, aaBB.

50)The process of transfer of genetic information from DNA to RNA/formation of RNA from DNA is called_____?

(a) transversion(b) transcription(c) translation(d) translocation.

51) **A** DNA with unequal nitrogen bases would most probably be_____?
(a) single stranded(b) double stranded(c) both of these (d) none of these

52) When Sex is determined in human beings?
(a) before fertilization (b) at time of fertilization(c) 40 days after fertilization(d) seventh to eight week when genitals differentiate in foetus.

53) Each chromosome at the anaphase stage of a bone marrow cell in our body has_____?
(a) two chromatids(b) several chromatids(c) no chromatids(d) only one chromatid

54)The Place on chromosome where spindle fibers are attached during cell division is called_____?

a) chromatids b) centromere c) centrioles d) mitochondria

55) Chromosome doubling in every generation reproducing sexually is prevented by

A. meiosis
B. reduction division
C. both A and B
D. mitosis

56) An estimate measurement of a chromosome in close circle of Escherichia Coli is

A. 1400 µm
B. 1200 µm
C. 1100 µm
D. 1000 µm

57) There are 48 chromosomes in a cell of a

A. chimpanzee
B. human being
C. drosophila
D. monkey

58) What are the major chemical components of the chromosome?
A] DNA and RNA
B] DNA and Protiens
C] Protein and Histone
D] Protein and Nucleic acid

59) What is called genome?
A] The haploid set of chromosomes
B] Thedipploid set of chromosomes
C] The fusion of two haploid gametes
D] The somatic cells

60)how many autosomes in human beings?
A] 23 pair
B] 22 pair
C] 1 pair
D] 44 pair

61) . Chromosome's diameter vary from:
A] 0.1 to 50 µm
B] 0.2 to 2 µm
C] 1 to 500 µm
D] 2 to 20 µm

62) The DNA threads which appear inside the nucleus at the time of cell division
a) Spindle fibers
b) Centrioles
c) Asters
d) Chromosomes

63)chromatin is composed of
a) DNA
b) DNA and proteins
c) DNA, RNA and proteins
d) None

64) Which of the following histones bind to linker DNA?
a) H1
b) H2A
c) H2B
d) H3

65) The sister chromatids separate at
a) Prophase
b) Metaphase
c) Telophase
d) Anaphase

66) Chromosomes are arranged along cell equator to form

A. metaphase plate
B. anaphase plate
C. telophase plate
D. prophase plate

67) Number of chromosomes in parent cell are

A. equal
B. not equal
C. 38 chromosomes
D. 48 chromosomes

68) Chromosomes are divided in anaphase to form

A. daughter chromosomes
B. sister chromosomes
C. daughter chromatids
D. sister chromatids

69) Kinetochores is preset in each chromosome at

A. somatic cells
B. vacuoles
C. centromere
D. gametophytes

Chromosomes are composed of

A. 40% protein and 60% DNA

B. 50% protein and 50% DNA

C. 70% protein and 30% DNA

D. 60% protein and 40% DNA

70)

71) Region of repetitive nucleotide sequences at each end of chromatid is

A. centromere
B. telomere
C. isomers
D. Monomer

72) If centromere is located at terminal end of chromosome than its called

A. metacentric
B. submetacentric
C. acrocentric
D. telocentric

73) f length of chromosome arms are unequal than chromosome is said to be

A. metacentric
B. submetacentric
C. acrocentric
D. telocentric

74) For vertebrates, sequence of nucleotides in telomeres is

A. GGGTAA
B. AAAGTT
C. TTTAGG
D. TTAGGG

75) Tightly packed form of DNA is called

A. supercoiling
B. compressed state

C. Euchromatin
D. Heterochromatin

76) Type of chromatin that participates in active transcription of DNA to mRNA products is

A. heterochromatin
B. Euchromatin
C. Centromere
D. Acrocentric chromosomes

Answer

77) Heterochromatin is usually localized to periphery of the

A. cell membrane
B. cytoplasm
C. nucleus
D. cell wall

78) Euchromatin is the

A. lightly packed form of chromatin
B. tightly packed form of chromatin
C. concentrated form of chromatin
D. elongated form of chromatin

Answer

**79)**Percentage of human genome to be euchromatic is

A. 0.72
B. 0.52
C. 0.92
D. 0.5

80) When molecule relieves helical stress by twisting around itself than phenomena is called

A. coiling
B. supercoiling

C. elongation
D. compression

81) are the fluorescent bands observed after quinacrine mustard staining with UV light.

A)Q bands

B)RBandin

C)C Banding

82)The nucleosome is composed of a core of eight histone proteins and DNA wrapped around them.

A) 145bp

B) 140 bp

C) 142bp

D) 147 bp

83)WRITE THE NAME OF Nucleosomes- Five histones?

ANS;Five abundant histones are H1, H2A, H2B, H3 and H4.

 The core histones share a common structural fold, called histone-fold domain

84)IN which the karyotype of a species may be represented diagrammatically, showing all the morphological features of the chromosome; such a diagram is known as Idiotype.

a) idiotype

b) karyotype

c)Kinetochore

d)none of these

85)define CEN DNA . write the name of sub domain of Kinetochore?

Ans:The DNA sequence within these regions is called CEN DNA ..120 base pairs long and consists of sub-domains, CDE-I, CDE-II and CDE-III.

86) n prokaryotes single copy DNA is present

A. 40% per genome
B. 30% per genome
C. 20% per genome
D. 10% per genome

Answer

**87)**In most eukaryotic organisms telomerase is active only in

A. germ cells
B. embryonic stem cells
C. certain white blood cells
D. all of above

Answer

**88).** Percentage of human genome to be euchromatic is

A. 0.72
B. 0.52
C. 0.92
D. 0.5

89)The number of gene may be as high as in some larger bacteriophage genome. ?

a) 150

b)160

c)170

d) 180

90) XY set of chromosomes are termed as

A. Allosomes
B. autosomes
C. anosomes
D. hetrosomes

**91)** In humans, males have set of chromosome

A. XX
B. XY
C. YX
D. YY

92) Considering Deoxyribonucleic Acid (DNA) structure, backbone outside double helix is made up of

A. sugar and nitrogen
B. nitrogen and carbon
C. phosphate and sugar
D. phosphate and nitrogen

**93)**Considering Deoxyribonucleic Acid (DNA) structure, inside basis of double helix is made up of

A. nitrogen
B. oxygen
C. phosphorus
D. magnesium

**94)**Deoxyribonucleic Acid (DNA) structure was proposed in

A. 1949
B. 1963
C. 1958
D. 1953

**95)**Scientists who proposed Deoxyribonucleic Acid (DNA) structures are

A. Francis Crick and James Watson
B. James Crick and Francis Watson

C. Marie Curie and Niels Bohr
D. Nikola Tesla and Marie Curie

**96)**Sequence of amino acids (R-CH($NH_2$)-COOH) of Deoxyribonucleic Acid (DNA) is controlled by sequence of

A. dominant proteins
B. nucleosomes
C. nucleotides
D. chromatin

97) Two strands of DNA is a double helix structure are coiled with each other

A. parallel
B. ant parallel
C. opposite
D. connected

98)Giant chromosomes are made dark staining regions called

A) bands"

B)Beads

C)Dark band

d)"interband

99) In Drosophila about bands can be recognized.

a) 6000 bands

b)700bands

c) 5000 bands

d)1000bands

100) in Giant chromosome Found in of some Diptera.

a) salivary glands

b)parathyroids.

c)adrenal glands.

d)all of these

101)Lampbrush chromosome First observed by ?

a) Flemming in 1883

b)Flemming in 1882

c)Flemming in 1885

d) Flemming in 1888

102)Lampbrush chromosome ae found in ?

a)vertebrates

b)invertebrates

c)all of these

**103)**Which one of the following best defines binary fission?

a) The process by which one cell splits into two cells

b) The process by which one cell splits into many cells

c) The process by which two cells split into many cells

d) The process by which many pairs of cells reproduce

104) The binary fission process includes the following steps: 1)The cell grows larger, 2)The cell undergoes cytokinesis and 3)Two daughter cells are produced. What step is missing?

a) Antibiotic resistance is transferred.

b) DNA is copied.

c) The cell undergoes conjugation.

d) All of the answers are correct.

105) What is asexual reproduction?

a) A type of conjugation when an identical cell is made

b) A way to create genetic diversity

c) When an identical cell is made

d) When non-identical cells are made

106)

DNA is synthesized and chromosome number is doubled in

**A.** G1 phase **B.** S phase

**C.** G2 phase **D.** G0 phase

107)

In case of neurons, post mitotic cell escapes cell cycle and remain in _____ phase whthout proliferating further

**A.** G0 phase **B.** G1 phase

**C.** G2 phase **D.** S phase

108)

Mitosis occurs in

**A.** Diploid cells only **B.** Haploid cells only

**C.** Both diploid and haploid **D.** Monoploid cells

cells

109)

Microtubules are composed of

**A.** Action protein

**B.** Actin and myosin proteins

**C.** Sphingomyelin and tubulin

**D.** Tubulin protein and traces of RNA

110)

From each pair of centrioles _____ sets of microtubules originate

**A.** 2

**B.** 3

**C.** 4

**D.** 5

111)

Mitotic apparatus in animals is formed by

**A.** Aster and spindle

**B.** Polar and kinetochore microtubules

**C.** Spindle and RNA

**D.** Spindle only

112)

A special area of centromere with specific base arrangement and specific proteins where spindle fibers are attached is called

**A.** Chromatin

**B.** Aster

**C.** Kinetochore **D.** Primary constriction

113)

Each kinetochore gets _____ fibers in mitosis

**A.** 5 **B.** 4

**C.** 3 **D.** 2

114)

Period between two consecutive divisions is called

**A.** Interphase **B.** Resting phase

**C.** Mitotic phase **D.** Cell cycle

115)

Most critical phase of mitosis is

**A.** Prophase **B.** Anaphase

**C.** Metaphase **D.** Telophase

116)

Reverse of prophase is

**A.** Interphase **B.** Metaphase

**C.** Anaphase **D.** Telophase

117)

In plants mitotic apparatus consists of

**A.** Centriole **B.** Centriole and asters

**C.** Centrioles, asters and spindles **D.** Spindles only

118)

Meiosis occurs at the time of

**A.** Gamete formation in animals **B.** Spore formattion in plants

**C.** Growth in animals **D.** Both a and b

119)

Interphase of meiosis lacks

**A.** G0 phase **B.** G1 phase

**C.** G2 phase **D.** S phase

120)

Homologous chromosomes are

| | | | |
|---|---|---|---|
| **A.** | Similar and identical | **B.** | Identical but not similar |
| **C.** | Not similar and not identical | **D.** | Similar but not identical |

121). The longest phase of meiosis I is

| | | | |
|---|---|---|---|
| **A.** | Metaphase | **B.** | Anaphase |
| **C.** | Prophase | **D.** | Telophase |

122. Tetrad or bivalent is

| | | | |
|---|---|---|---|
| **A.** | Paired homologous chromosome but not fused complex structure | **B.** | 2 unpaired homologous chromosome |
| **C.** | Paired and fused complex structure | **D.** | None of these |

123). Pairing of homologous chromosomes is completed in

| | | | |
|---|---|---|---|
| **A.** | Leptotene | **B.** | Zygotene |
| **C.** | Pachytene | **D.** | Diplotene |

124). Each bivalent has

| | | | |
|---|---|---|---|
| **A.** | 3 chromatids | **B.** | 4 chromatids |
| **C.** | 5 chromatids | **D.** | 2 chromatids |

125). Crossing over occurs between

| | | | |
|---|---|---|---|
| **A.** | Sister chromatids | **B.** | Non sister chromatids |
| **C.** | Independent chromatids | **D.** | Each and every chromatid |

126). Inability of chromosomes to segregate during anaphase and telophase of meiosis is called

| | | | |
|---|---|---|---|
| **A.** | Crossing over | **B.** | Chromosomal disjunction |
| **C.** | Chromosomal nondisjunction | **D.** | None of these |

127). Which statement is incorrect?

| | | | |
|---|---|---|---|
| **A.** | Down's syndrome involves autosome | **B.** | In Turner's syndrome 2n-1 |
| **C.** | In Klinefelter's syndrome XYY | **D.** | In Down's syndrome sperms are commonly involved |

128) Apoptosis

| | | | |
|---|---|---|---|
| **A.** | Cell death due to tissue damage | **B.** | Causes inflammation |
| **C.** | Internal programme of events by which cell commits suicide | **D.** | Damages neighboring cells |

129). A cell of human being has 46 chromosomes; it divides to form some daughter cells, each having 23 pairs of chromosomes. The division would be

| | | | |
|---|---|---|---|
| **A.** | Mitosis | **B.** | Meiosis |
| **C.** | Amitosis | **D.** | Budding |

130) Meiosis was discovered by

A. Oscar Hertwig
B. Walther Flemming
C. Charles Darwin
D. Robert H. Brown

131) rocess of cell division that results in two genetically identical daughter cells developing from a single parent cell is

A. cell division
B. mitosis
C. meiosis
D. binary division

132) Chromosome number in meiosis is

A. doubled
B. reduced by half
C. reduced by 2
D. increased by 3

133) Abnormal separation of chromosomes is classified as

A. diploid budding
B. haploid budding
C. non-disjunction
D. disjunction

134) Pair of homologous chromosomes is classified as

A. tetravalent
B. trivalent
C. bivalent
D. covalent

135) In anaphase I of meiosis, chromosomes contains one pair of

A. daughter nucleosomes
B. sister nucleosomes
C. daughter chromatids
D. sister chromatids

136) Period of rest in which daughter cells enters after meiosis I is

A. karyokinesis
B. cytokinesis
C. interkinesis
D. vacuokinesis

**1)Write the Sub Types of Genetics?**

Ans: There are four sub disciplines of genetics which are as follows, although some of the Geneticist classify Genetics in many sub disciplines

- Transmission (Classical) Genetics
- Population Genetics
- Quantitative Genetics
- Molecular Genetics

2) **Gregor Mendel used garden pea as experimental plant for formulating the ?without option**

Ans: laws of heredity

3) **Gregor Mendel used garden pea as experimental plant for formulating the laws of heredity, write the following were the properties of garden pea ?**

Ans: Seed in a variety of shapes and colors.

Self, cross pollinate.

Takes up little space.□

Short generation time □

Produces many offspring

**4)Define Spermatogenesis and oogenesis ?**

Ans;Meiosis in males is called spermatogenesis and produces sperms.

Meiosis in females is called oogenesis and produces ova.

**5)name the stages of Prophase I of meiosis ?**

Ans;Prophase I occupies more than 90% of the time required for meiosis.
Five sub stages of prophase I

Sub stages of prophase I

- □ Leptotene
- □ Zygotene
- □ Pachytene
- □ Diplotene
- □ Diakinesis

**6)write the Experiment of T. H. Morgan** ?

ANS;Eye color trait is linked with sex of drosophila

- □ Morgan with an experiment proved that eye color trait is linked with sex of the drosophila.
- Experiment
- □ Male flies were with white eyes (mutant).
- □ Female flies were with red eyes (wild type).
- Progeny f1 was with red eyes
- □ All the flies produced during F1 generation were with red eyes.
- Progeny f2 showed 3:1 ratio
- □ F2 generation showed 3:1 (three red eye: one white eye).

- □ White-eyed trait appeared only in males.
- □ All females and half males with red eye.
- White eyed trait appeared only in males
- □ White-eyed trait appeared only in male fruit flies (Drosophila melanogaster).

Morgan's Discovery: Transmission of X chromosome correlates with inheritance of eye color trait First solid evidence indicating that a gene is associated with a specific chromosome

**7) Define Linkage map**

Ans; A linkage map is a genetic map of a species that shows the position of its known genes or genetic markers relative to each other in terms of recombination frequency.

**8)write the name of Markers used in linkage map;**

Ans; Microsatellites

□ Restriction fragment length polymorphisms (RFLPs).

**9)what are the the SRY gene - Y chromosome ?**

Ans; Researchers have found the SRY gene (sex determining region of the Y chromosome

- □ In individuals with the SRY gene, the generic embryonic gonads are modified into testes.Other genes on Y chromosome
- □ In addition, other genes on the Y chromosome are necessary for the production of functional sperms.
- □ In individuals lacking the SRY gene, the generic embryonic gonads develop into ovaries.

iotechnologists

**10)Human Sex linked Disorders ?**

Ans; Color blindness

□ Duchenne muscular dystrophy

 Hemophilia

**11)Define Genetic mapping in humans?**

Ans;Genetic mapping

 Genetic mapping - also called linkage mapping provides clues about which chromosome contains the gene and precisely where it lies on that chromosome.

**12)Types of Genetic markers?**

Ans;Several different types of genetic markers, the type most used on genetic maps is known as a microsatellite and SNPs.

**13)Define Mono-allelic, di-allelic and poly-allelic**

Ans;30% genes in humans are di-allelic - they exist in two forms.

 70% are mono-allelic; they exist only in one form.

 Few are poly-allelic.

**14)Define allele?**

Ans;An **allele** is one of the possible forms of a gene. Most genes have two **alleles**, a dominant **allele** and a recessive **allele**. If an organism is heterozygous for that trait, or possesses one of each **allele**, then the dominant trait is expressed. ... **Alleles** were first **defined** by Gregor Mendel in the law of segregation.

15)

Define **Mitosis** And **Meiosis?**

Ans ; **Mitosis** And **Meiosis**. Cells give rise to new cells through a process called cell division. ... In this case, each cell divides twice through a process called **meiosis**, resulting in four daughter cells that contain only half the number of chromosomes when compared to the parent, and are not necessarily identical.

**16)define Kinetochore ?**

Ans;Within the centromere region, …locations where spindle fibers attach which consist of DNA as well as protein.

 The actual location where the attachment occurs is called the kinetochore and is composed of both DNA and protein.

**17)Define chromatin ?**

Ans;The complexes between eukaryotic DNA and proteins are called chromatin.

**18)DeineEuchromatin and Heterochromatin?**

Ans; **Chromatin** is made up of DNA, protein, and RNA. There are two forms of chromatins in the interphase nucleus: **euchromatin and heterochromatin**. Cytologically, **heterochromatin** stains more intensely than **euchromatin**. ...**Heterochromatin** is the tightly packed form of **chromatin** often found at the periphery of the nucleus.

**19)write the Types of Chromosome?**

ans: There are two types of eukaryotic chromosomes:

 Autosomes

 Sex chromosomes

 Autosomes: Paired chromosomes with the same length, shape, centromere location, and genes.

Any chromosomes other than sex chromosomes are called as autosomes.

Sex Chromosomes: Members of a pair of chromosomes that differ between males and females.

**20)definePleiotropy ?**

Ans;The ability of a gene to affect an organism in many ways is called pleiotropy

**21)Define Epistasis ?**

Ans; Alleles at one gene locus can hide or prevent expression of alleles at second gene locus.

Example

 A dominant allele B for black, recessive allele b brown coat, a second gene (e) locus controls eumelanin deposited in fur.

 Dogs homozygous recessive at this locus (ee) will have yellow fur no matter which alleles are at first locus.

**22)define Polygenic traits ?**

Ans; Most traits are not controlled by a single gene locus, but by the combined interaction of many gene loci.

 These traits are called polygenic traits.

Example: Humans Height

**23)define Co-dominance ?**

Ans;A situation, when both alleles appear in the phenotype.

 Neither allele is dominant.

 Both alleles are expressed in a heterozygous individuals.

 Parallel behavior of both allele.

Example 1 : Blood group

**24)Types of variation:?**

Two types of Variation

Ans;Discontinuous Variation

 Continuous Variation

Discontinuous Variation

Discontinuous variations are entirely genetically controlled.

They cannot be altered by external conditions. Blood group, color blind, dwarfism are genetically controlled.

Continuous variation

Continuous variation describes the situation in which there are many intermediates between the extremes. Shade of hair color between black and blond.

**25)Define mutation?**

Ans; mutation: Indicate "a change" while in other disciplines it is used to indicate "a diseasecausing change". Mutations may be harmful, beneficial, may have no effect.

**26)Write the note on prophase ;metaphase .Anaphase telophase;**

Prophase of mitosis: Prophase is characterized by four events: Chromosomes condensed - more visible. Nuclear membrane disappears. Centrioles are separated and take positions on the opposite poles of the cell. Spindle fibers form and radiate toward the center of the cell.

 Chromosomes condense at the start of mitosis.

 DNA wraps around histones that condense it.

Prophase- chromosomes are condensed- Different levels of condensation/folding Chromosomes condense at the start of mitosis. DNA wraps around histones that condense it.

Prophase Summary Prophase is the first stage of mitotic cell division.

Metaphase During metaphase, two events occurs; Chromosomes line up across the middle of the cell. Spindle fibers connect the centromere of each sister chromatid to the poles of the cell, where centrioles are present. Metaphase – chromosomes are aligned in the middle Metaphase-Summary During metaphase, two events occurs; Chromosomes lined up in the middle line. Spindle fibers connect to centromere.

Anaphase of mitosis During anaphase, three events occurs; Centromere split – as a result sister chromatids are separated. Sister chromatids become individual chromosomes. Separated chromatids move to opposite poles of the cell.

 Anaphase Securin is a protein which inhibits a protease known as separase. Destruction of securin results in split of centromeres and the new daughter chromosomes are pulled toward the poles. Cell in an oval shape Chromosomes are drawn to each side of the cell. Non-kinetochore spindle fibers push against each other,that stretch the cell into an oval shape

Summary Centromere split and chromosomes are separated. Chromosomes arrived at poles. Cell stretches into an oval shape. Telophase

consists of four events: Chromosomes start to uncoil. Nuclear envelope forms around the chromosomes at each pole of the cell.

Telophase Spindle fibers break down and dissolve, Daughter nucleus is formed in each of the daughter cell. Telophase Nuclear membranes are formed around each set of chromatids. Nucleoli also reappear. Chromosomes unwind back into chromatin material.

Telophase – effects of prophase are reversed Duringtelophase, the effects of prophase and prometaphase are reversed.

Telophase Summary Spindle fibers break down and nuclear members develops around each daughter nucleus.

**27)write the Modern Genetics Concepts ?**

Ans:  Chromosomes and Gene (s)

 Mendelian Genetics and Non-Mendelian Genetics

 Genotypes and Phenotypes 

Mutations and Polymorphism

**28) What is chromosome?**

Ans;

Genetic material in cells is organized into chromosomes. Prokaryotes generally have one circular chromosome. Eukaryotes - linear chromosomes in their nuclei.

**29) what is gene ?**

Ans; A heredity unit on chromosomes

**30) What is Trait?**

Ans variation for a character (i.e. brown, black, white etc).

**31)Define Genotype and Phenotype?**

: Genetic make-up of an organism is called Genotype while physical appearance of an organism is called Phenotype

**32)Define Dominant and Recessive?**

when one characteristic expresses itself over the other i.e. round over wrinkled was dominant in Gregor Mendel experiments while the trait that does not show through in the first generation is called as recessive trait i.e. wrinkled.